

AI Interview Coach

AI-Powered Resume Analysis & Interview Preparation

An intelligent platform that analyzes resumes, identifies skill gaps, generates interview questions, evaluates answers, and measures interview readiness — all powered by AI.

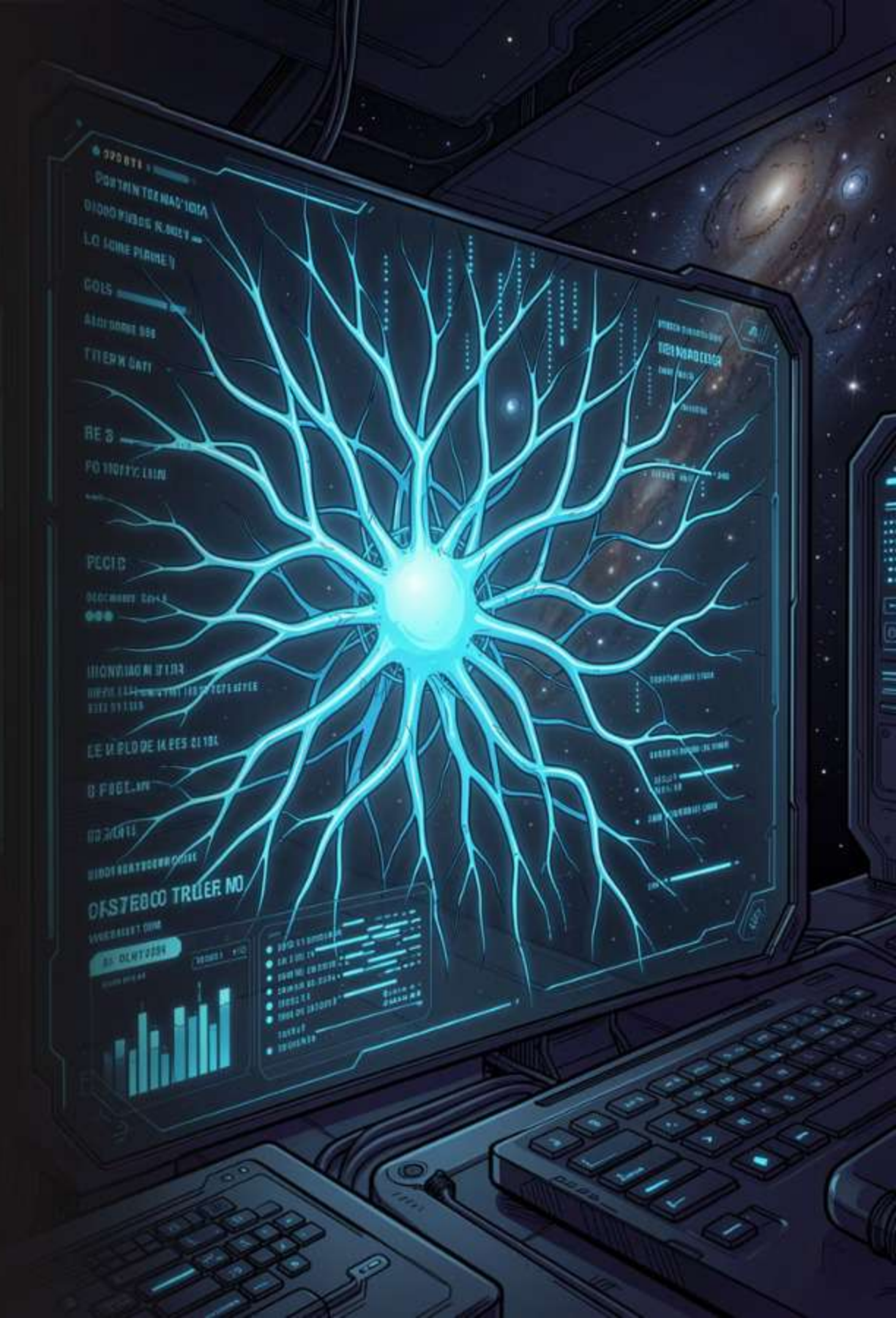
</> PYTHON

STREAMLIT

NLP & ML

PLOTLY

Presented by: Uday Deshmukh



The Problem: Job Seekers Are Underprepared

Most candidates fail interviews not because they lack talent — but because they lack targeted, intelligent preparation.

No Personalization

Generic prep materials ignore individual skill gaps and target roles.

Blind Spots

Candidates can't objectively identify what they're missing before an interview.

No Feedback Loop

Practice answers go unevaluated — no way to measure improvement.

Low Confidence

Uncertainty leads to poor performance and reduced placement success.



One Platform. Complete Interview Readiness.

AI Interview Coach replaces guesswork with data-driven preparation — from resume upload to readiness score.



→ Analyze Resumes

Parse PDFs and extract skills automatically using NLP.

→ Match Roles & Find Gaps

Compare candidate profiles against target job requirements.

→ Generate & Evaluate

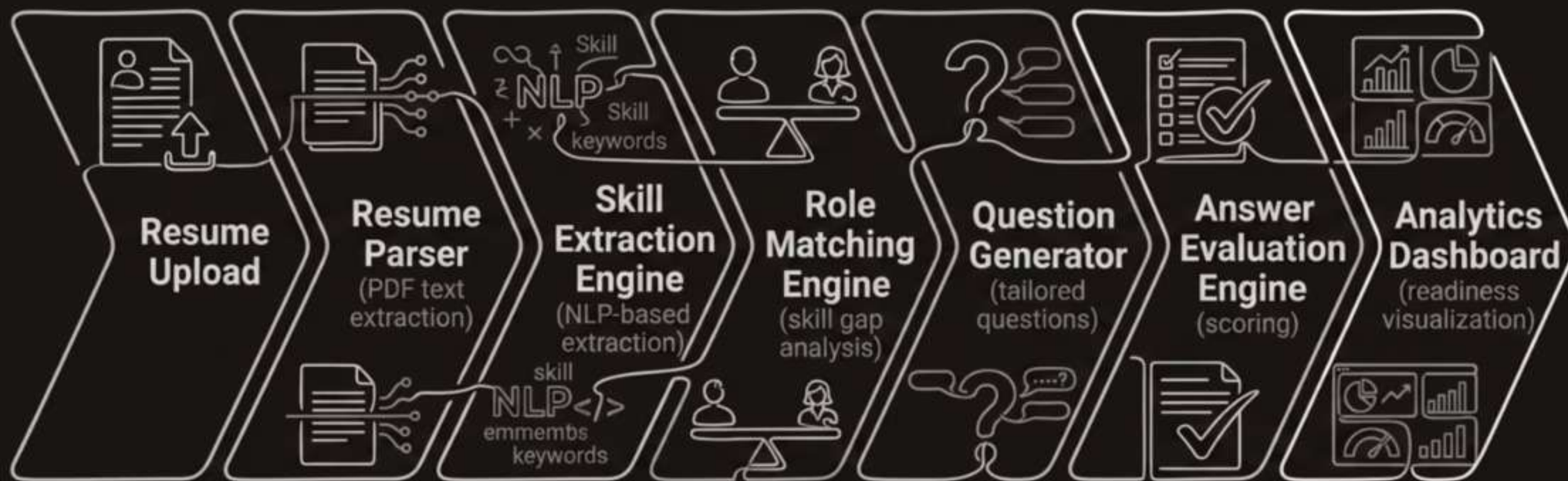
Produce tailored questions and score answers objectively.

→ Measure Readiness

Get a composite readiness score with clear classification.

End-to-End System Architecture

Seven intelligent stages transform a raw resume into a complete interview readiness profile — automatically and in seconds.



Technology Stack



Frontend (Streamlit)

Create fast, interactive web apps.



NLP & AI (spaCy, Sentence Transformers, NLTK)

Extract insights from text.



Data Processing (Pandas, NumPy)

Manipulate and analyze complex data.



Visualization (Plotly)

Generate interactive charts and figures.



Deployment (GitHub, Hugging Face Spaces)

Share models and host applications.

Built for Intelligence & Scale

Every layer of the stack is chosen for AI-native capabilities and rapid deployment.

- **Streamlit** — Rapid, interactive frontend for data apps
- **spaCy & Sentence Transformers** — Semantic NLP and embeddings
- **Pandas & NumPy** — High-performance data processing
- **Plotly** — Interactive, publication-ready visualizations
- **Hugging Face Spaces** — Zero-config cloud deployment

SKILL E STORCE

SKILL

DKIATFLOON

SKILL

SKILL 14

SKILL 15

AI ISKAIE

DKONWOLE

AIUOAGICTIUS

EXAREACTIOAS

VKFODCRIONS

 **Resume Parsing**

Upload any PDF resume. The engine extracts text, normalizes formatting, and structures data for downstream analysis.

 **Skill Extraction**

NLP models identify and categorize technical skills — Python, SQL, ML, AWS, Docker, Power BI — with frequency scoring.

A line graph with an upward-sloping arrow, indicating a positive trend or growth.

Skill Frequency Analysis

Highlights the most prominent skills and surfaces patterns to guide targeted preparation focus.

Role Matching & Question Generation

Role Matching Engine

Supported roles: Data Analyst, Data Scientist, Business Analyst, ML Engineer, Data Engineer, AI Engineer.

Match Score

Percentage fit for target role

Interview Question Generator

Questions are dynamically generated based on the candidate's resume and target role.

Skill Gap Report

Matched vs. missing skills

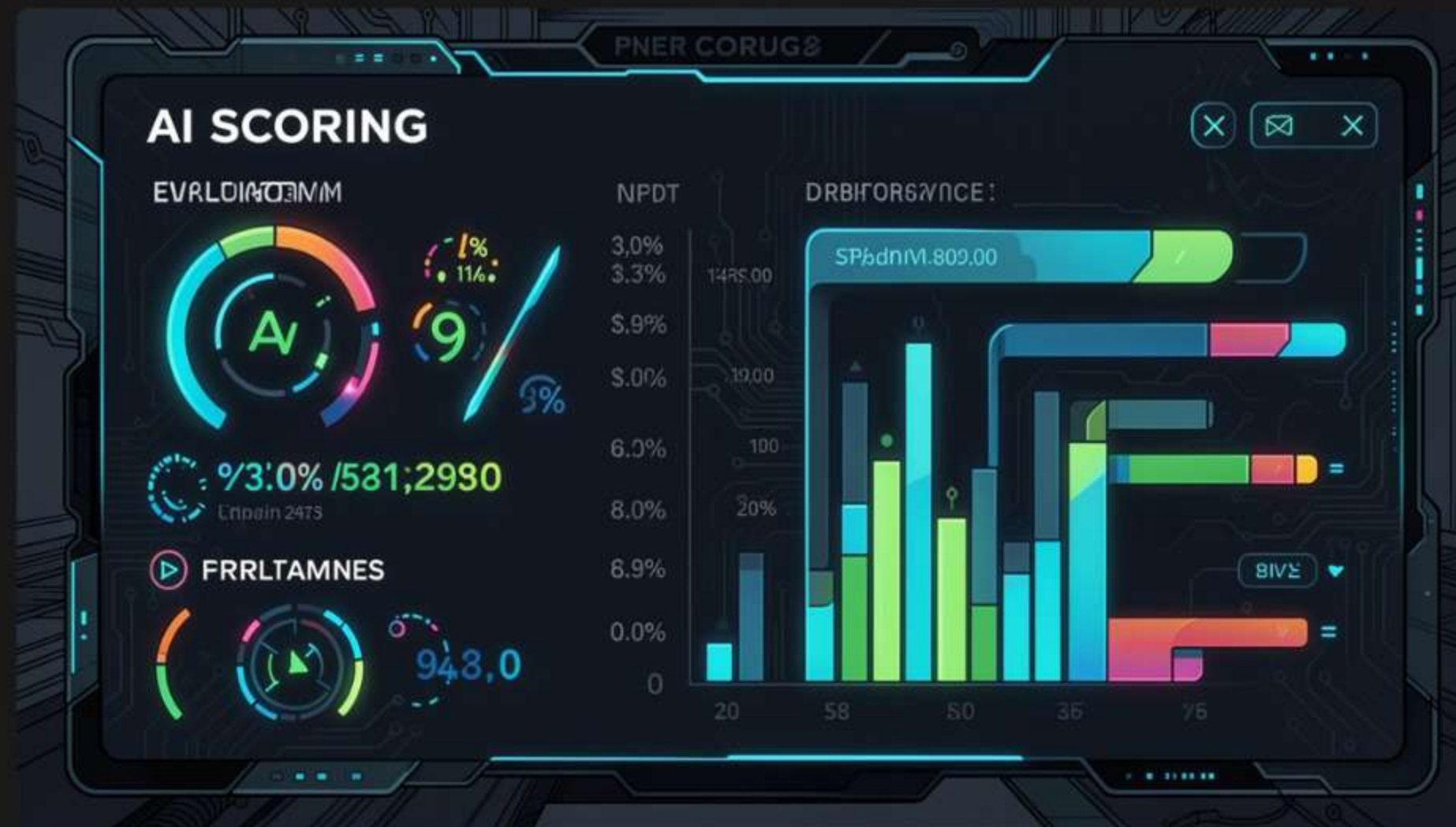
Difficulty

Easy · Medium · Hard

Categories

Technical · Behavioral · Scenario-Based

Answer Evaluation & Readiness Score



AI-Based Answer Evaluation

Every answer is scored across four dimensions:

- Technical Accuracy
- Communication Clarity
- Completeness
- Overall Performance

Readiness Classification: ● Advanced
● Intermediate ● Beginner —
based on resume match, skill coverage,
and question performance.

Interactive Analytics Dashboard

A real-time, visual command center for interview preparedness — built with Plotly.



KPI Cards

Match Score, Readiness Score, and Skill Coverage at a glance.



Radar & Gauge Charts

Visualize skill distribution and performance across dimensions.



Performance Analytics

Track improvement over time and identify areas to focus on next.



Business Impact & What's Next

Who Benefits

Students & Candidates

Personalized prep, faster upskilling, improved confidence.

Colleges & Placement Cells

Higher placement rates and measurable student outcomes.

Organizations & Recruiters

Better-screened, interview-ready candidates.

Roadmap

01

Voice-Based Mock Interviews

02

LLM-Powered Detailed Feedback

03

ATS Resume Optimization

04

AI Interview Simulation

05

Job Recommendation Engine

Thank You

AI Interview Coach — Empowering Candidates Through AI
Questions & Discussion Welcome

✉ UDAY DESHMUKH